#14

CS 0447
Introduction to
Computer Programming

The single-cycle CPU

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Luís Oliveira

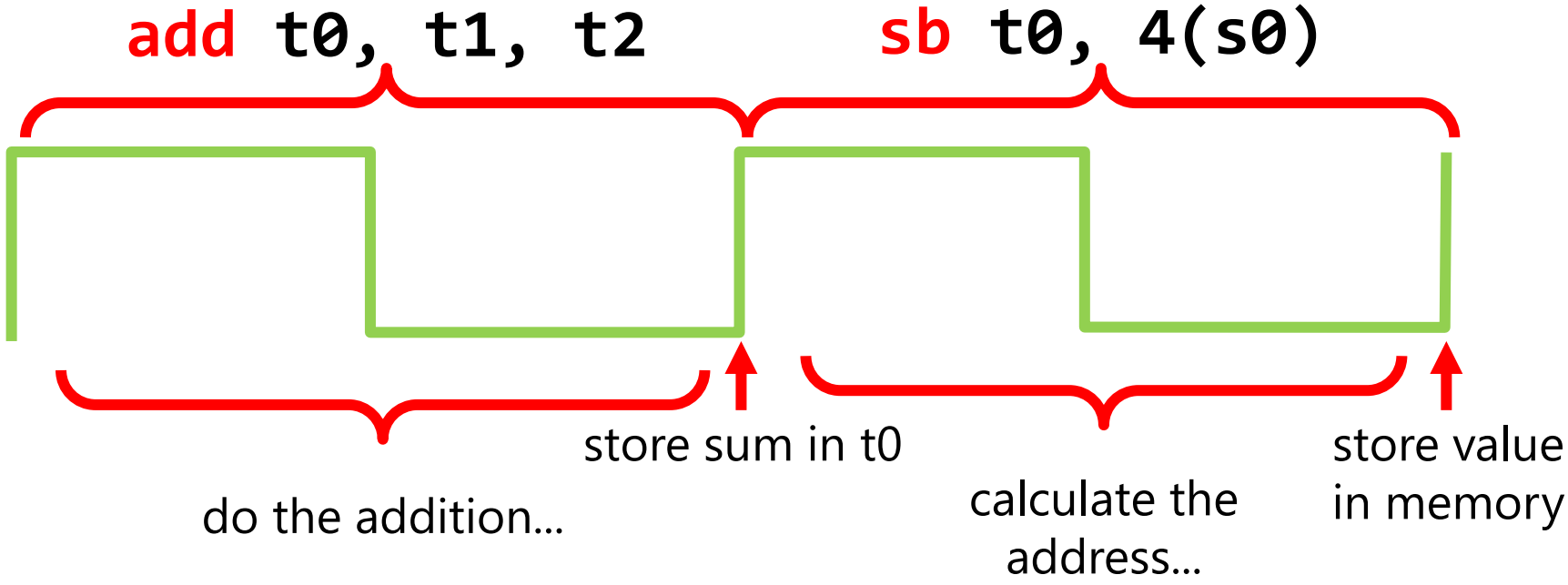
Fall 2020

Class announcements

- OMETs don’t forget them!

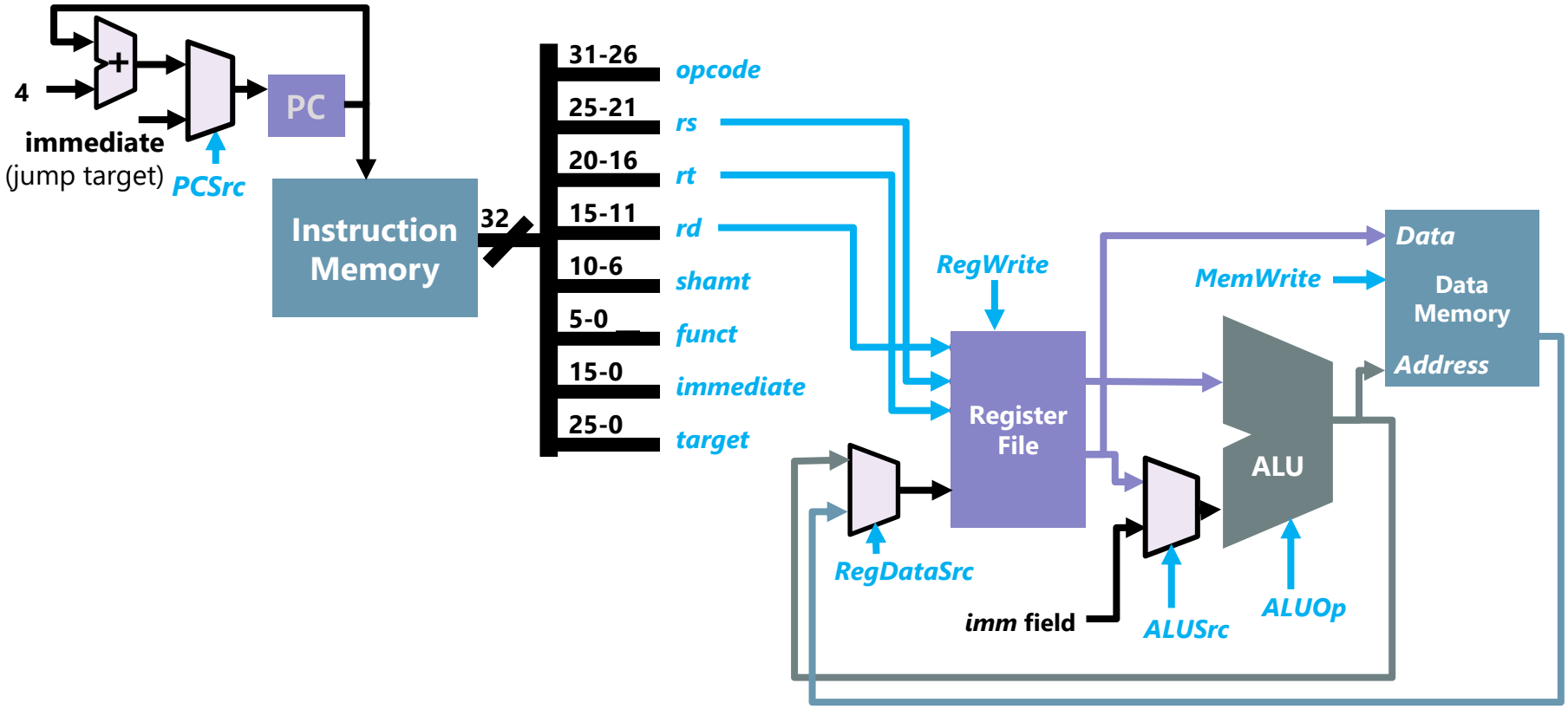
The single-cycle machine

- we've been talking about a **single-cycle** machine and building one for project 3
- this means each instruction takes **one clock cycle to execute**

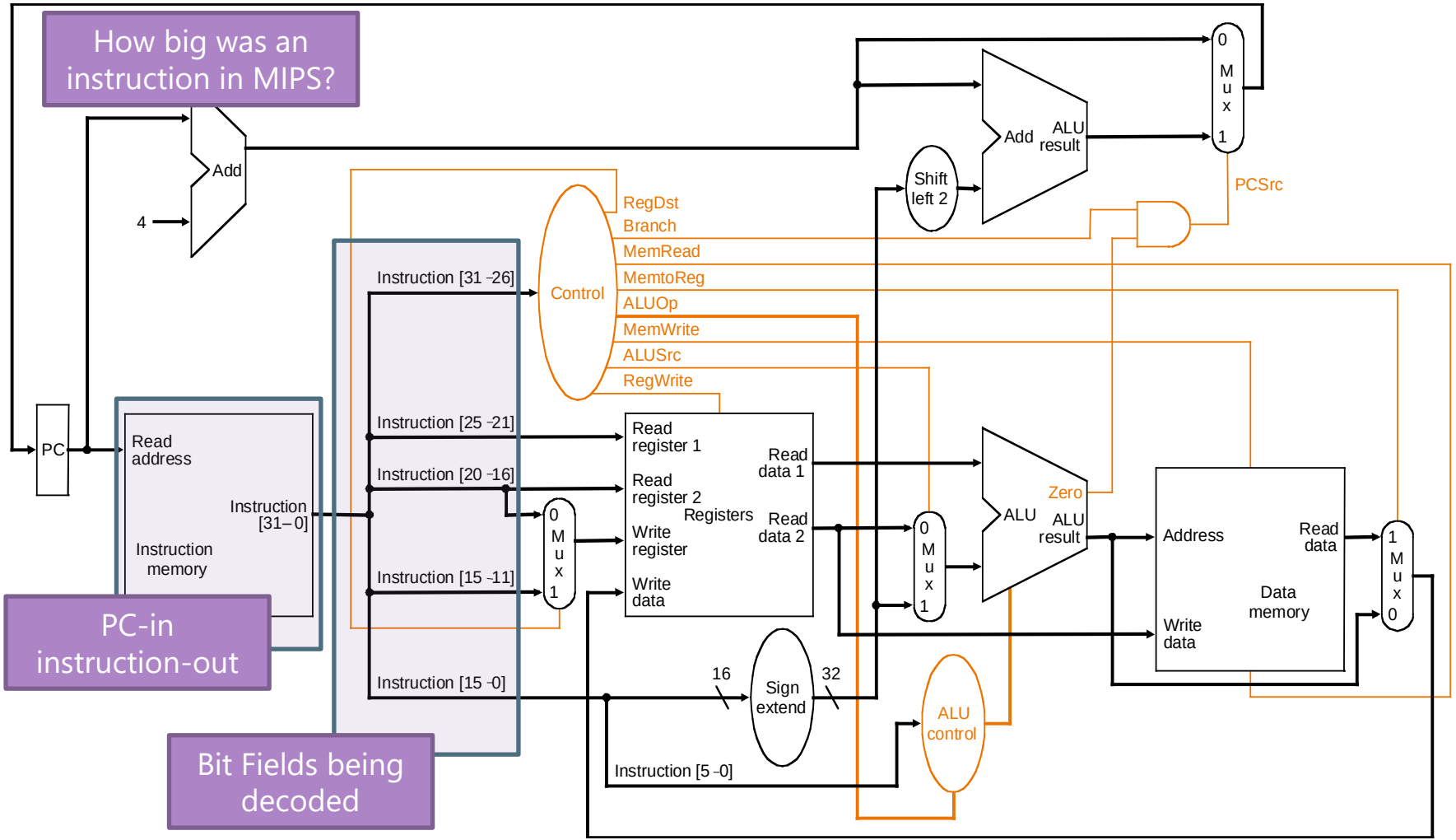


In this example each instruction **ends on the rising edge of the clock** since that's when the registers store their values

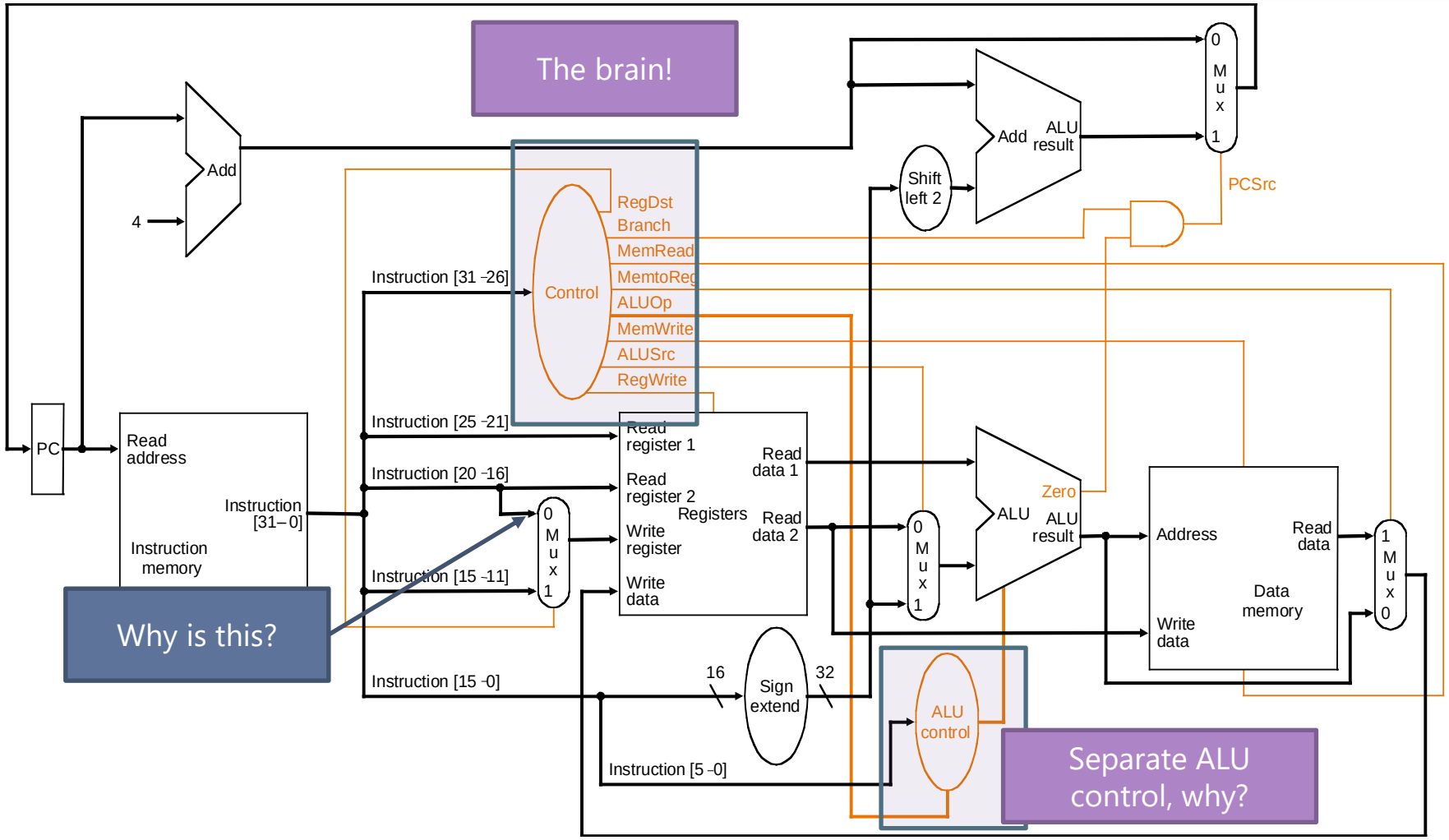
A global view



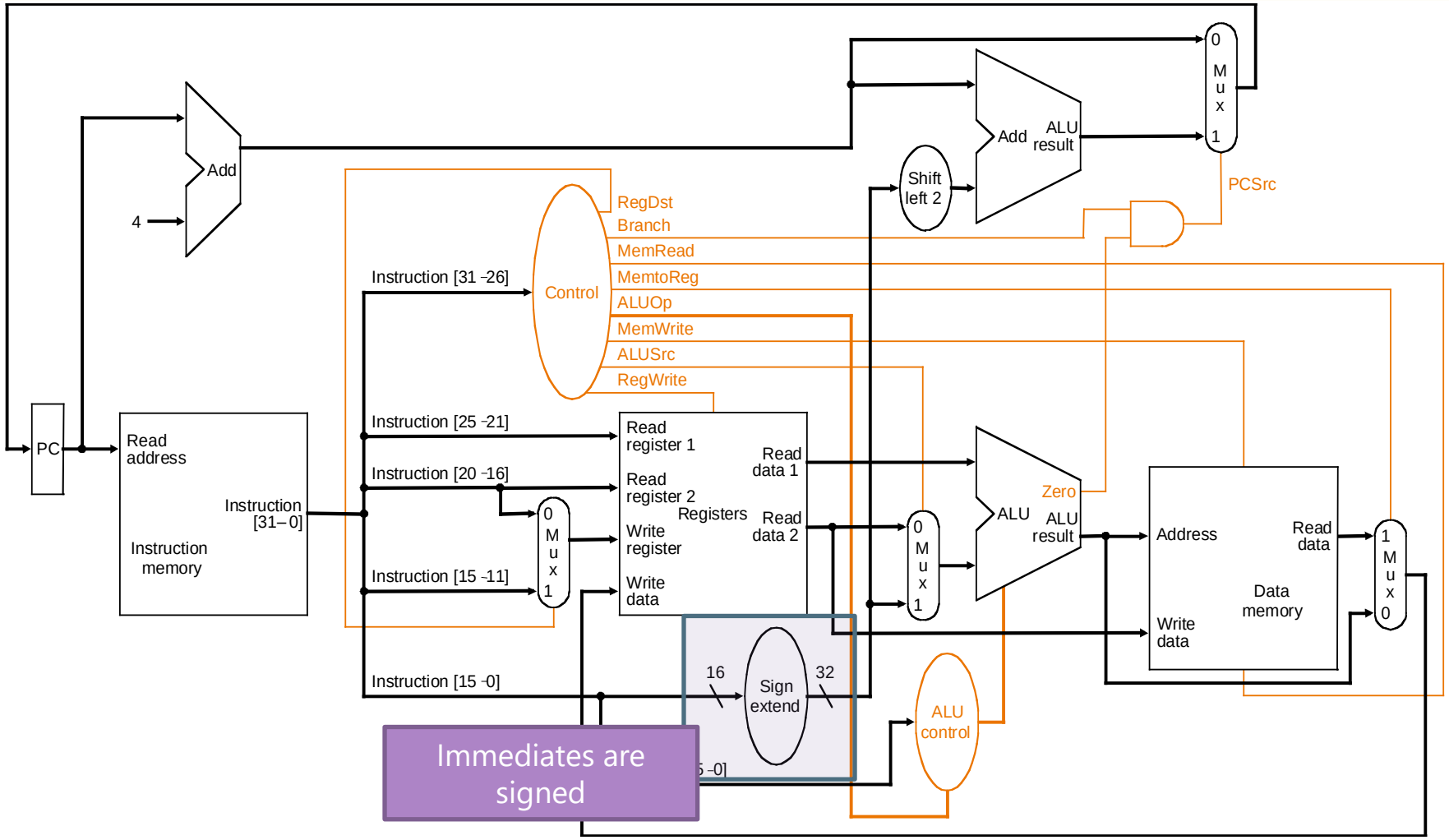
A global view



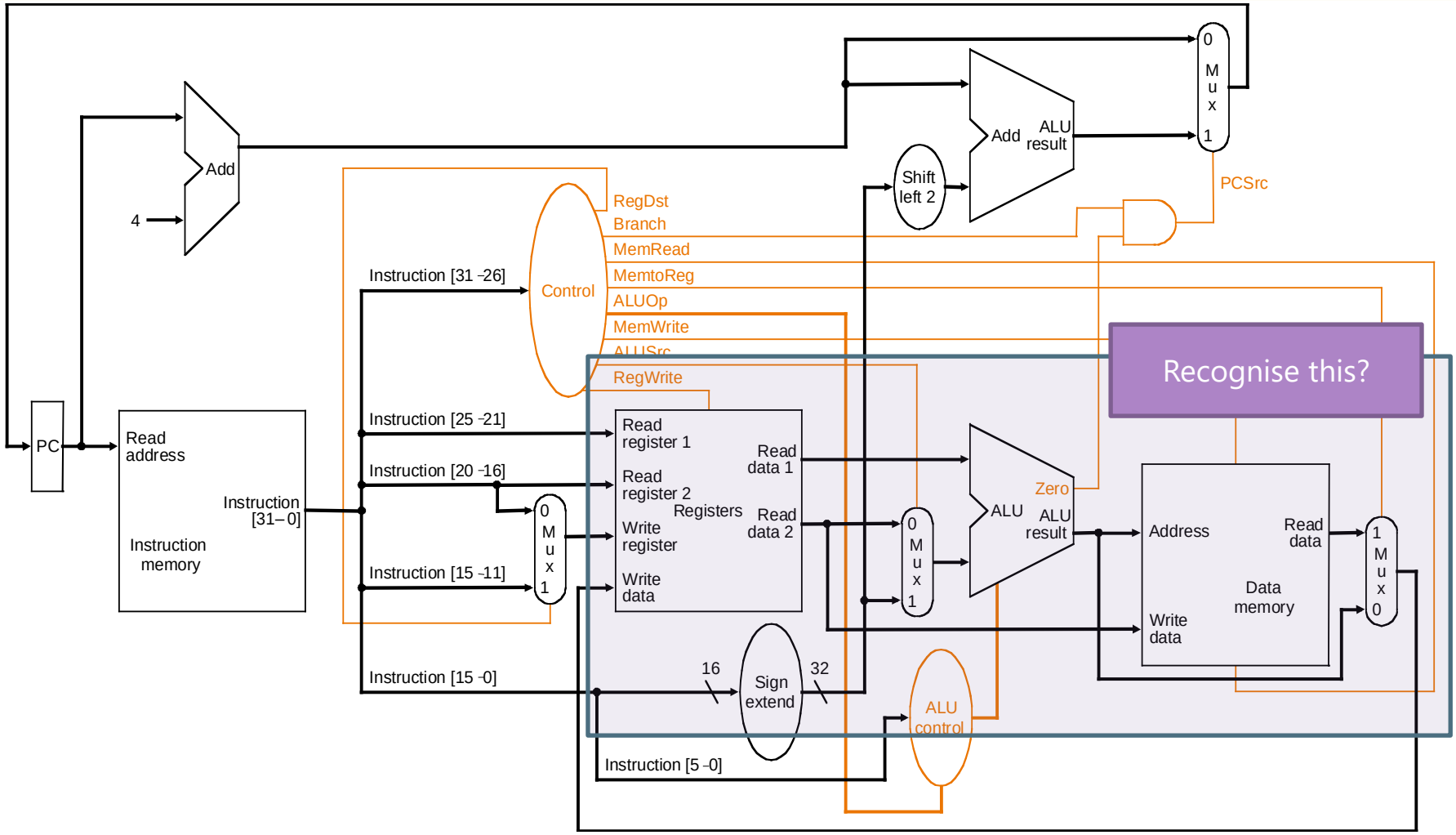
A global view



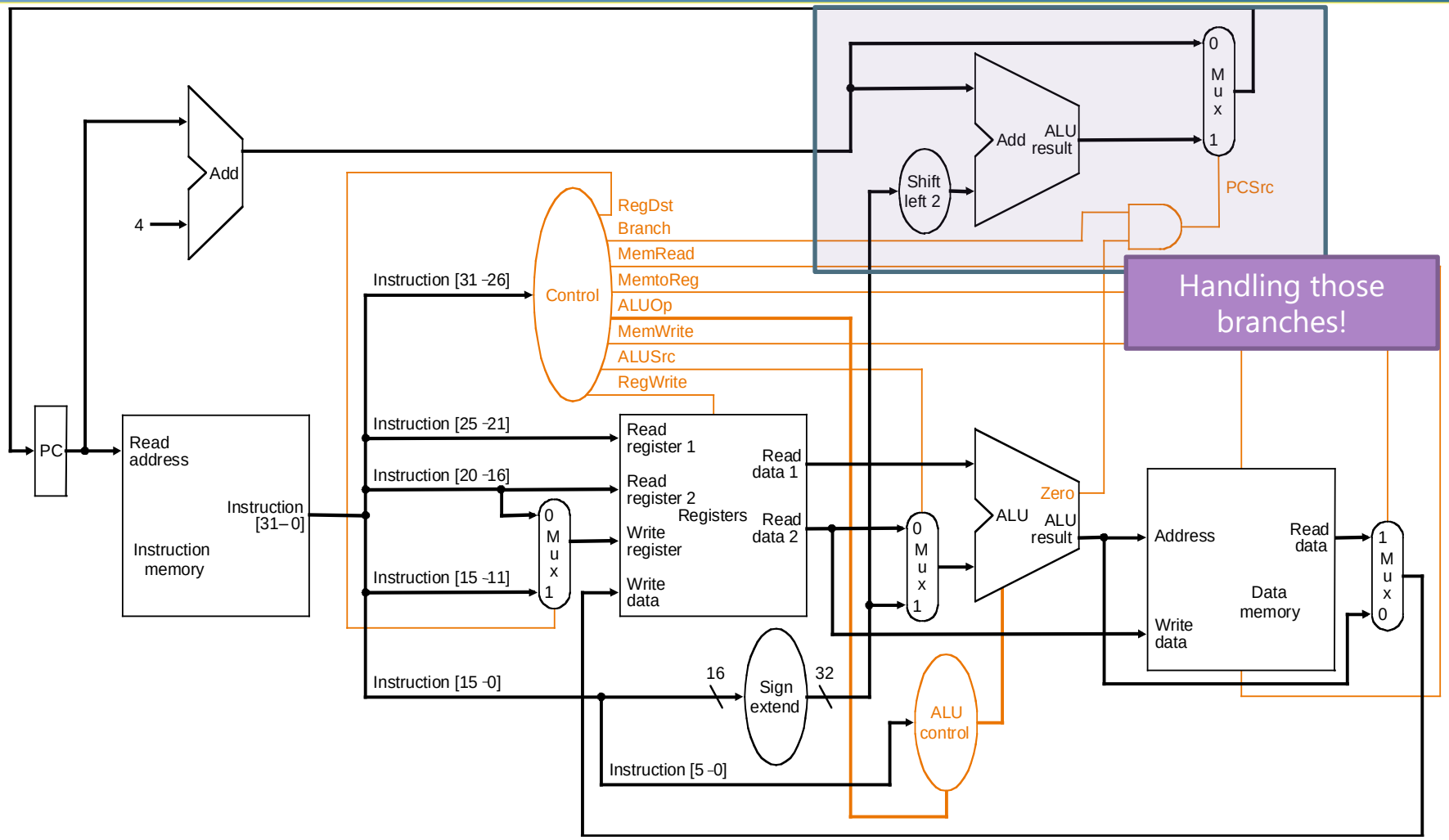
A global view



A global view

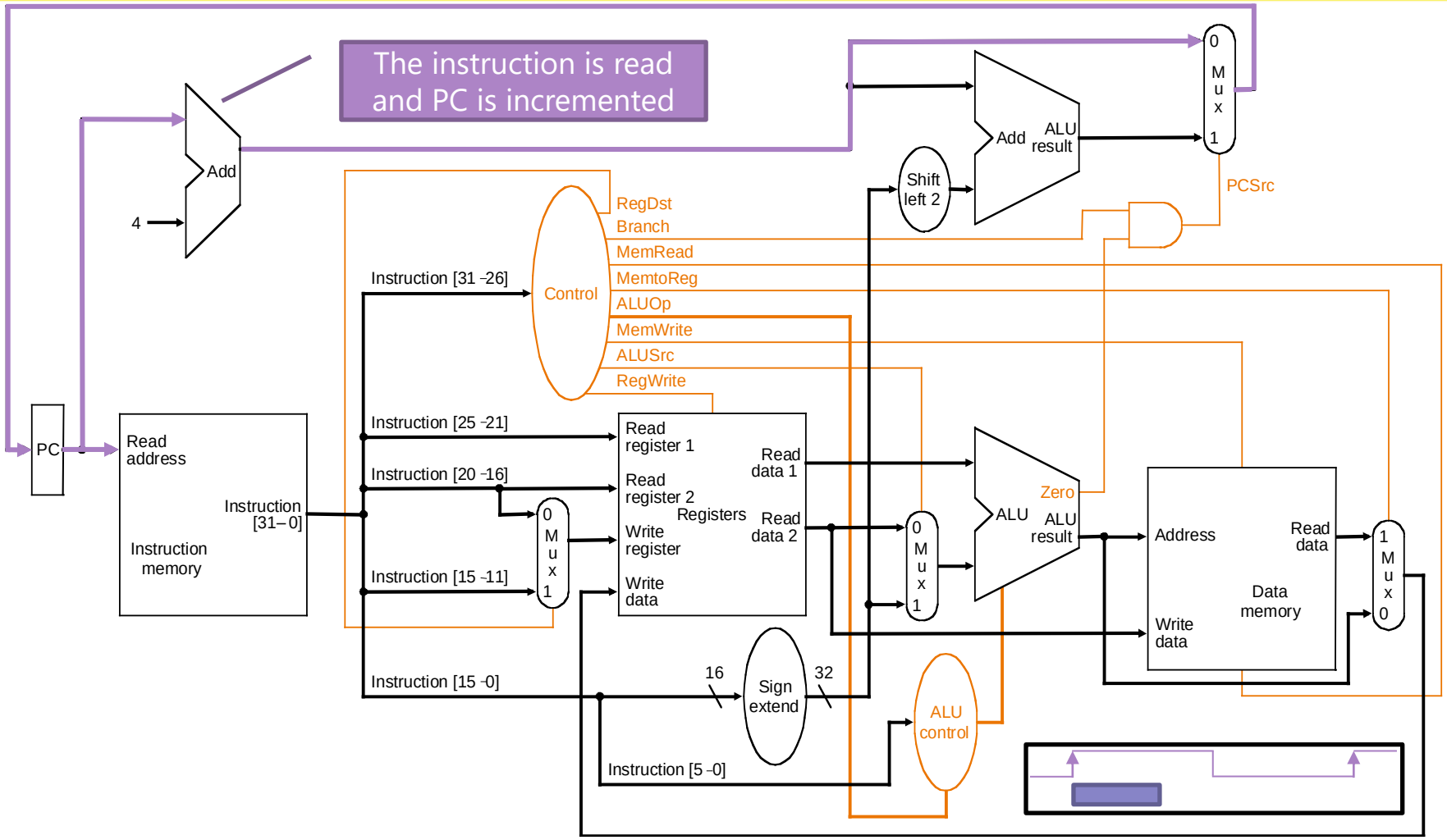


A global view

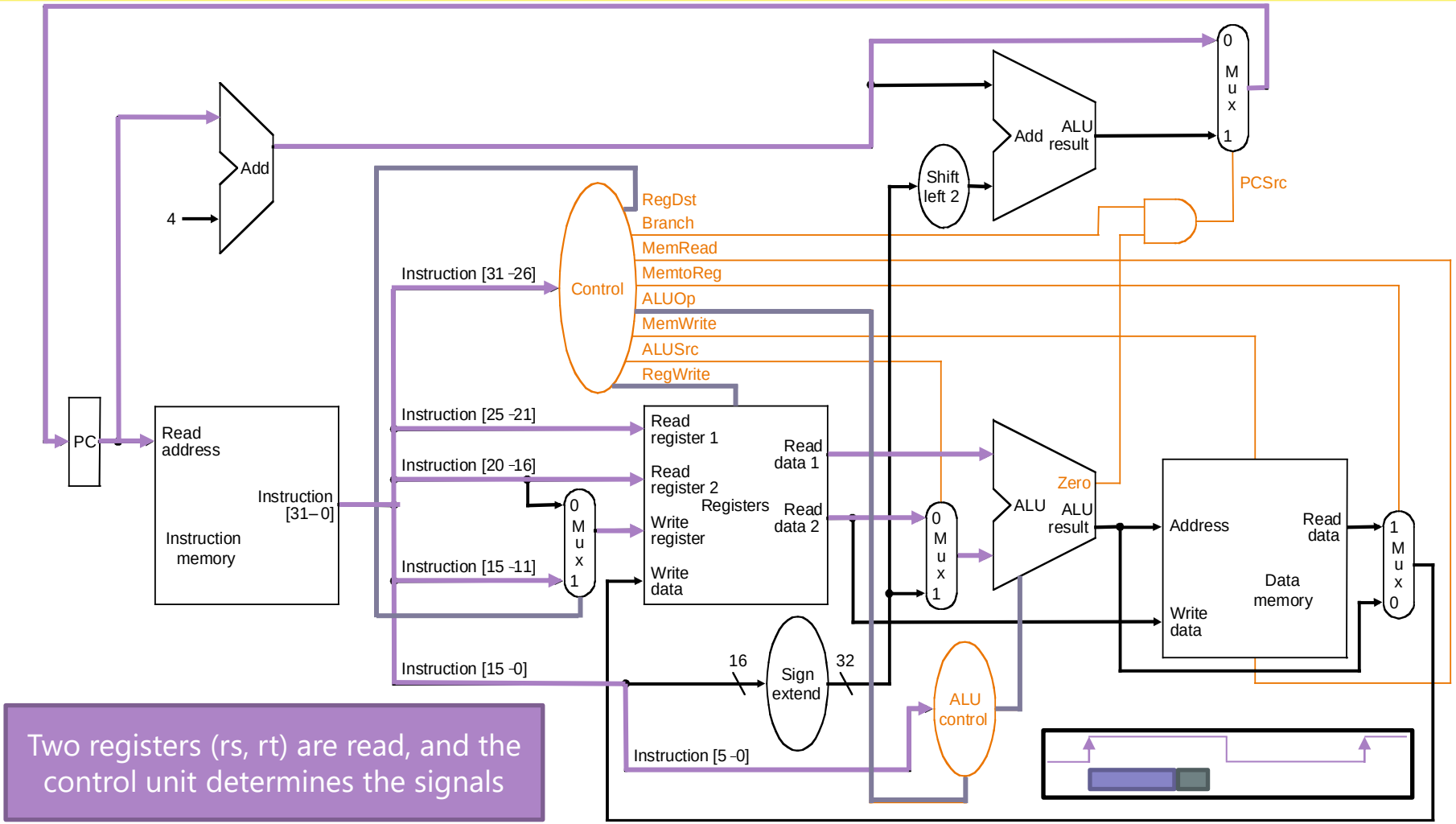


Executing an r-type instruction

Executing an r-type

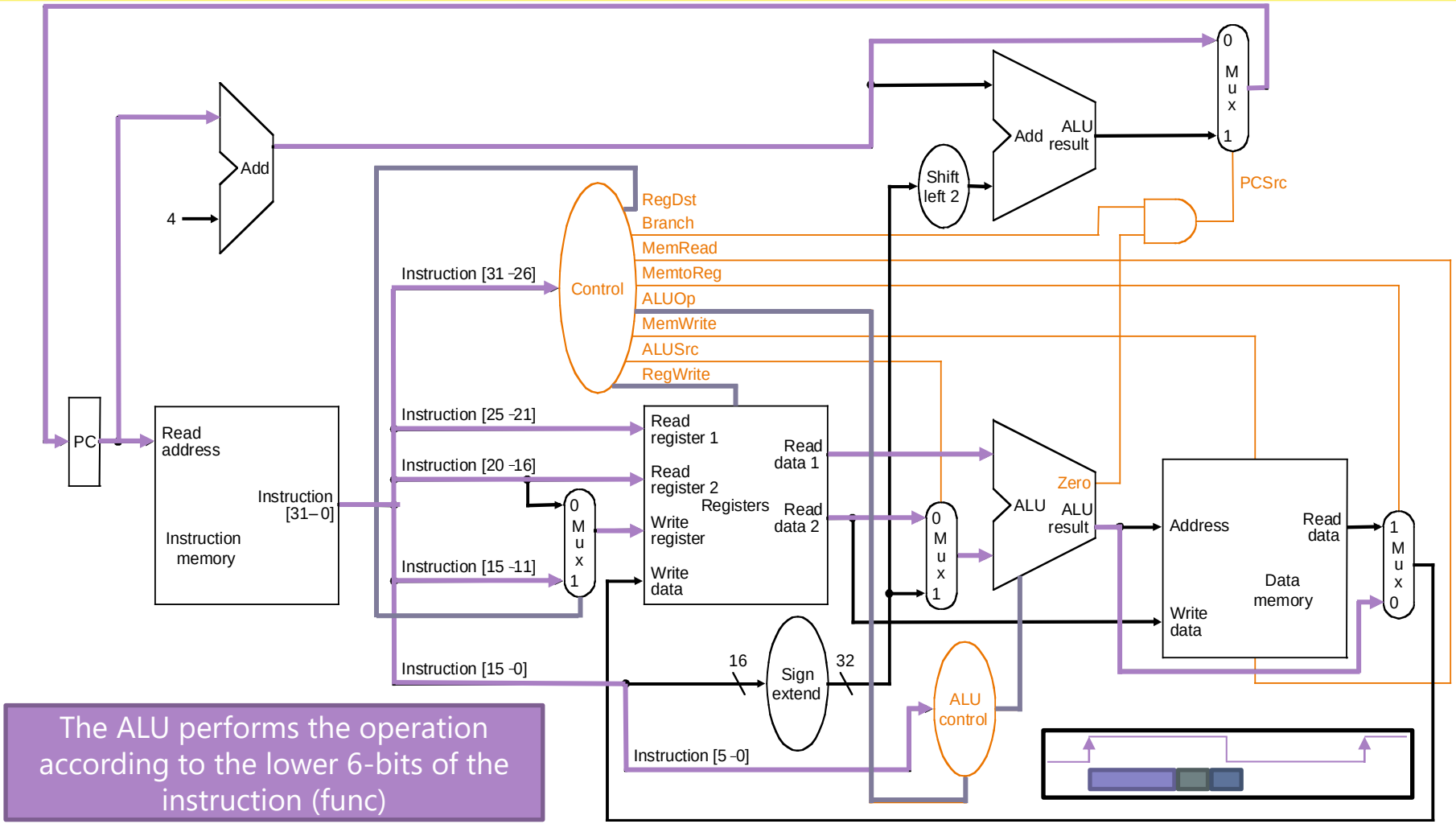


Executing an r-type



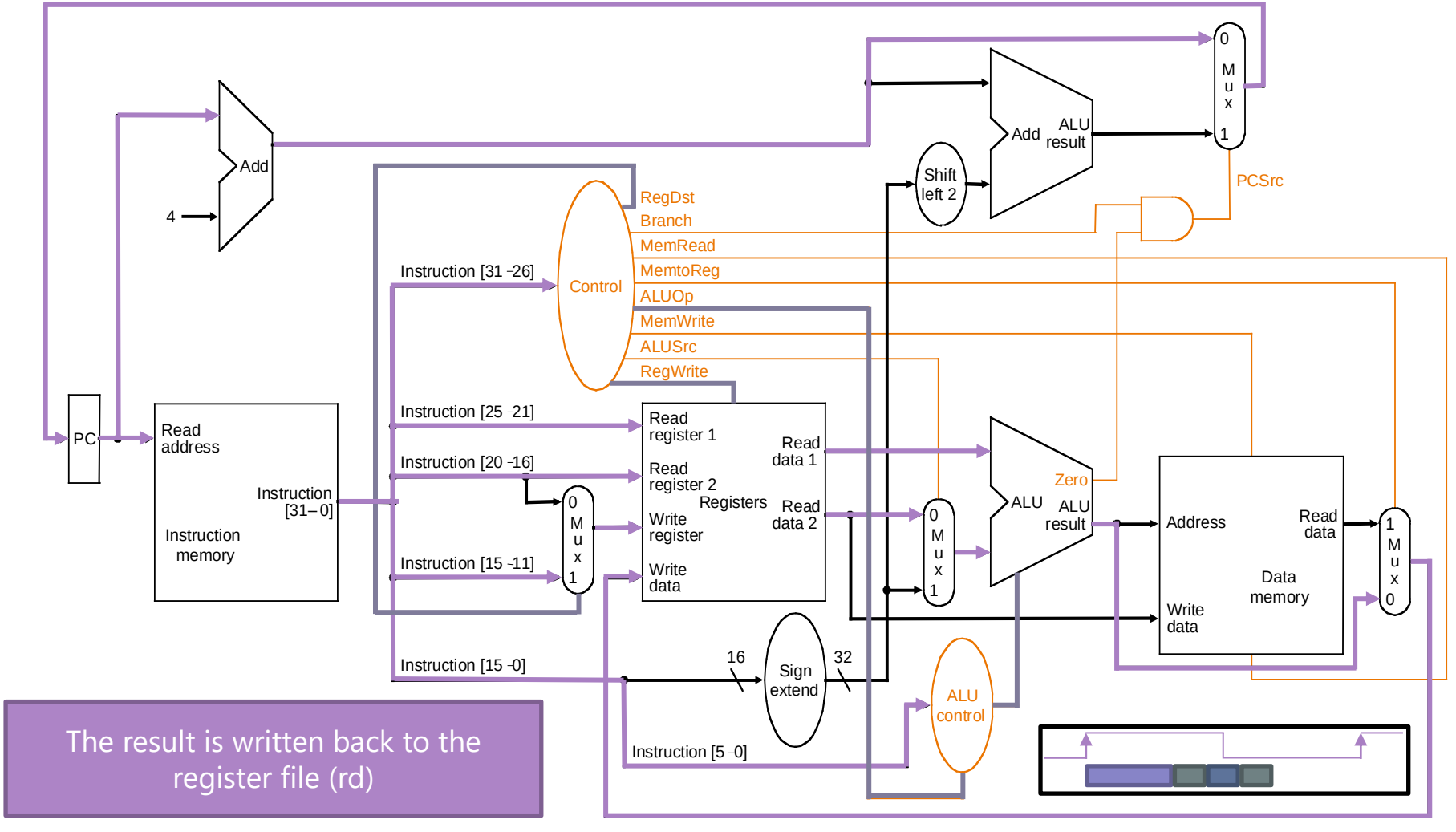
Two registers (rs, rt) are read, and the control unit determines the signals

Executing an r-type



The ALU performs the operation according to the lower 6-bits of the instruction (func)

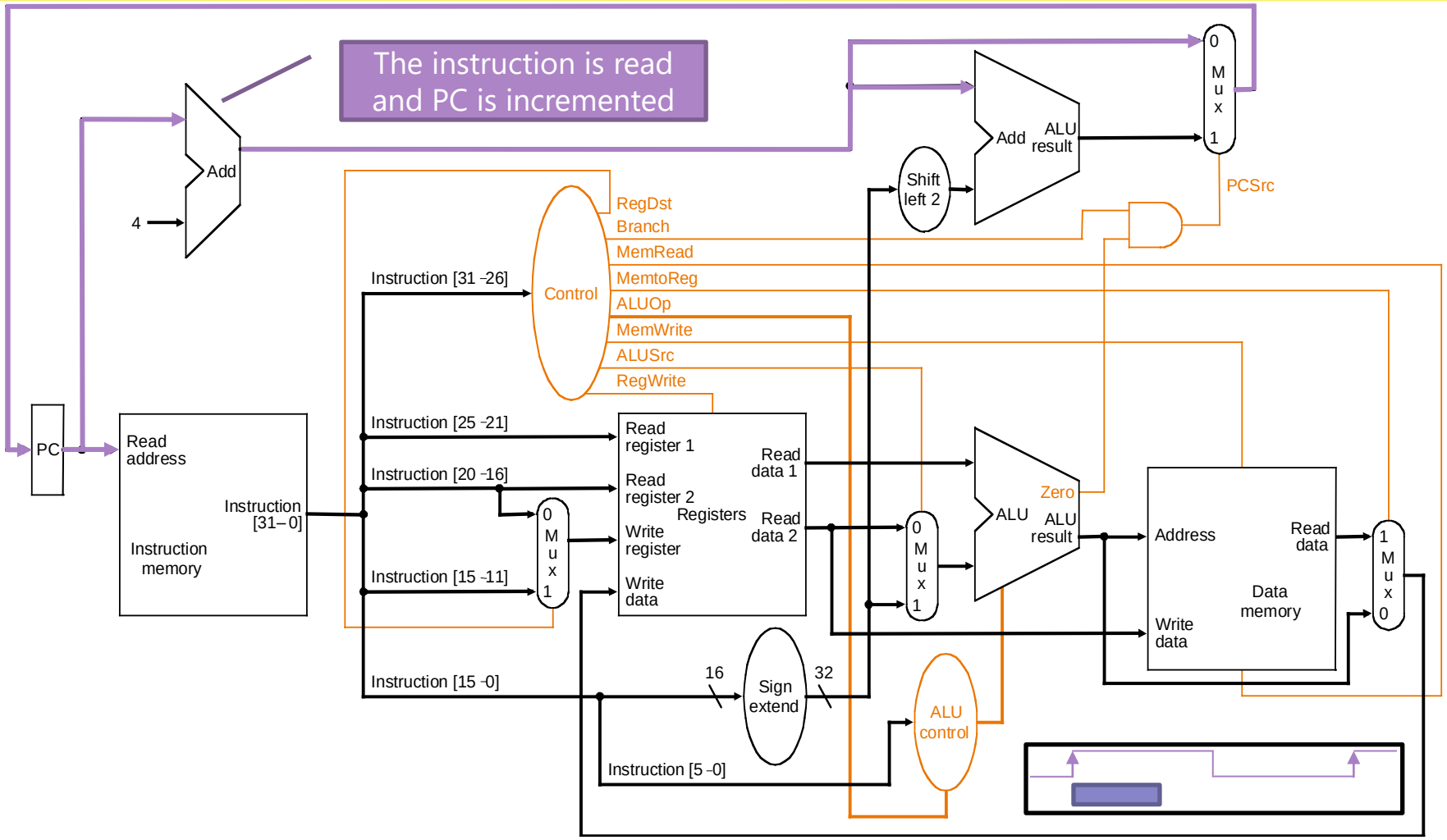
Executing an r-type



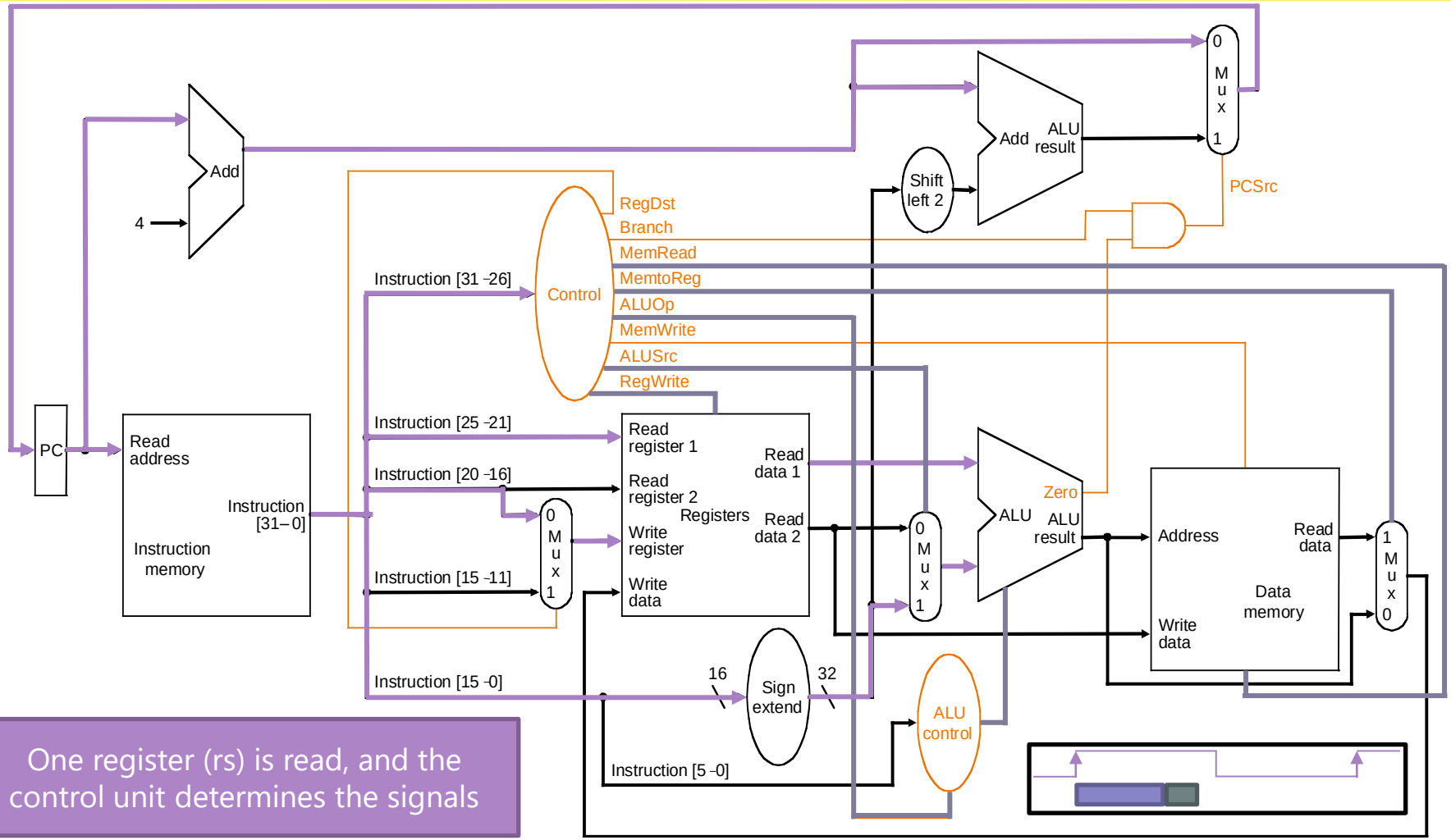
The result is written back to the register file (rd)

Executing a lw instruction

Executing an i-type : e.g., lw

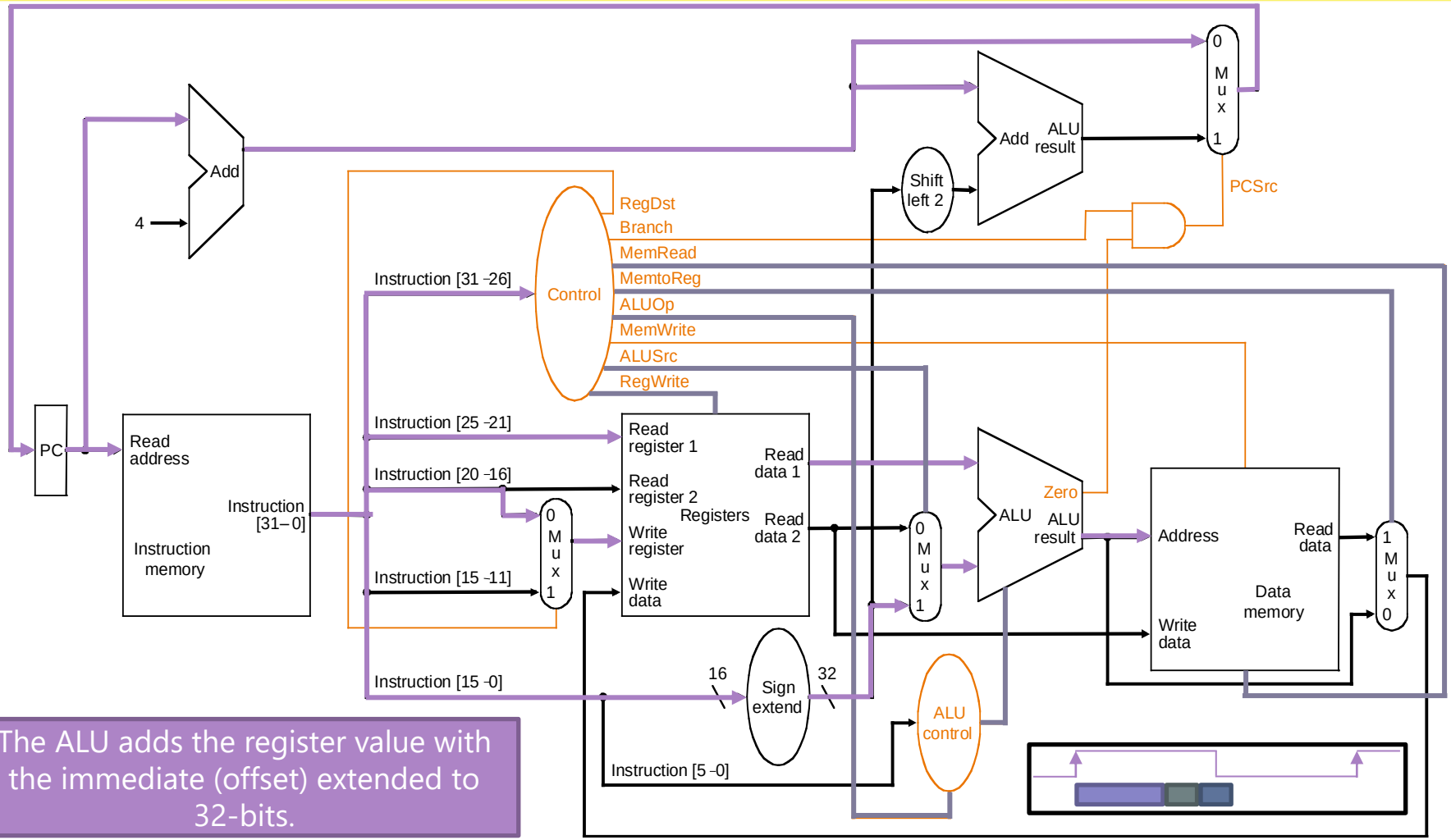


Executing an i-type : e.g., lw



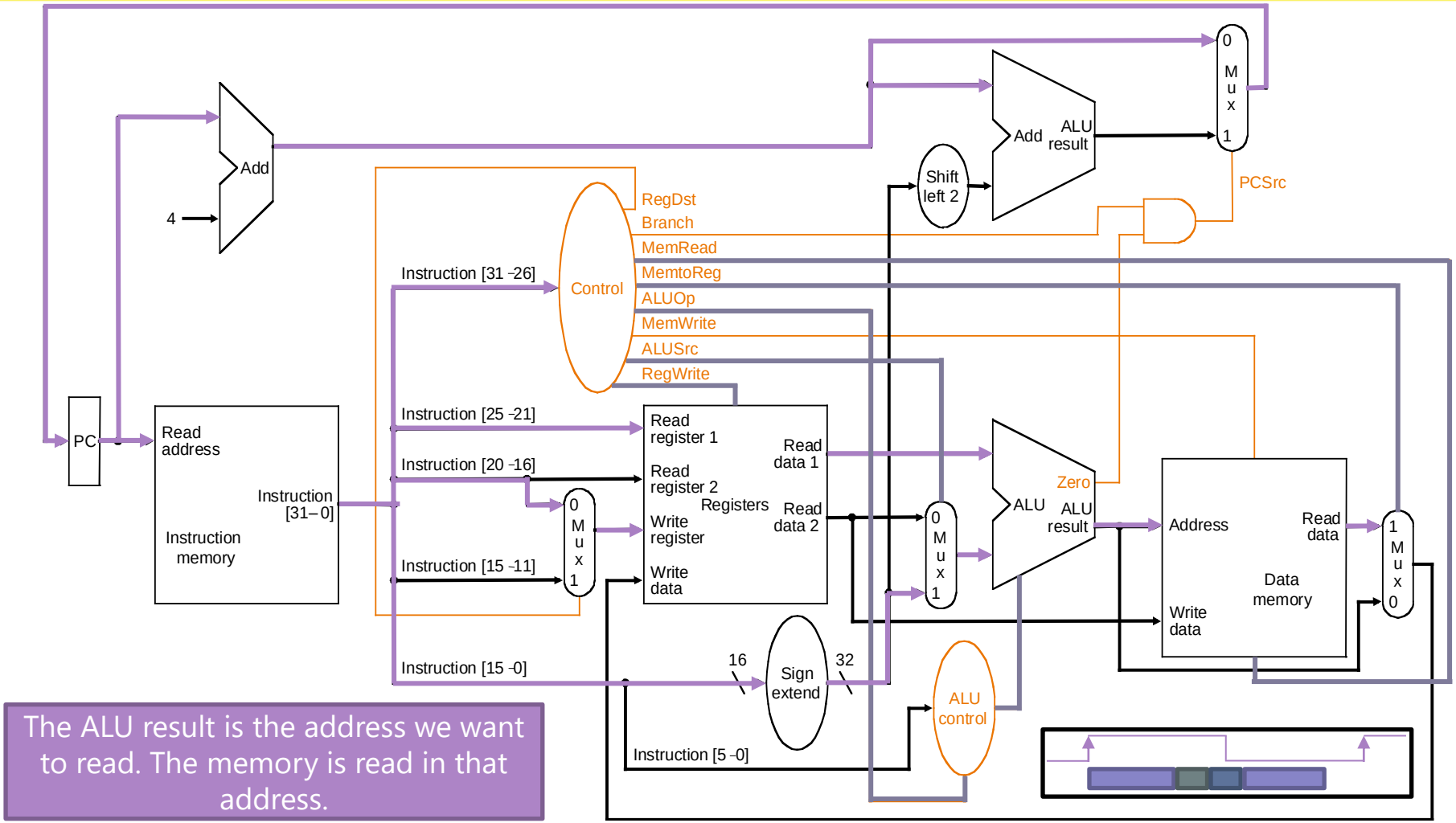
One register (rs) is read, and the control unit determines the signals

Executing an i-type : e.g., lw



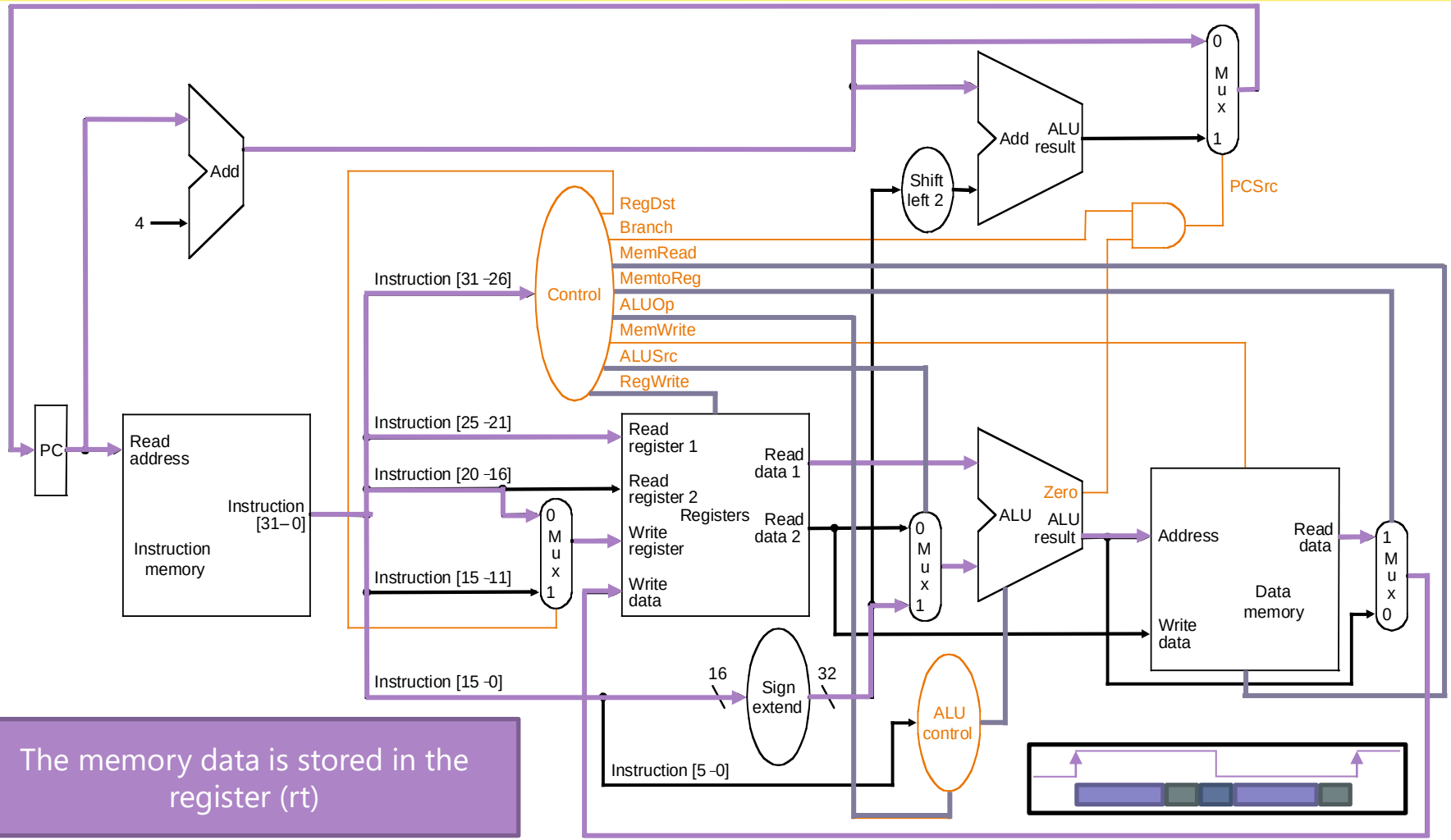
The ALU adds the register value with the immediate (offset) extended to 32-bits.

Executing an i-type : e.g., lw



The ALU result is the address we want to read. The memory is read in that address.

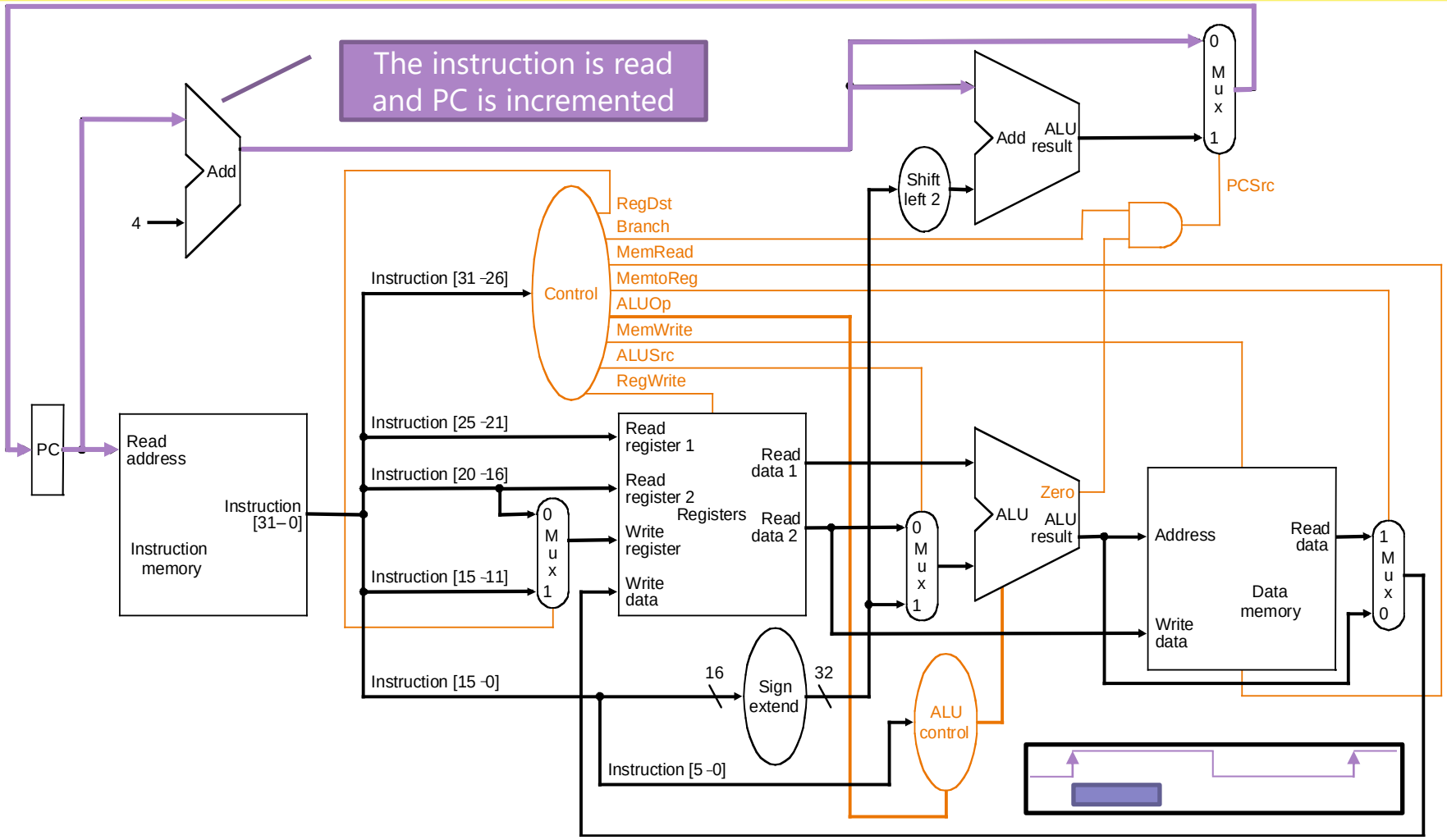
Executing an i-type : e.g., lw



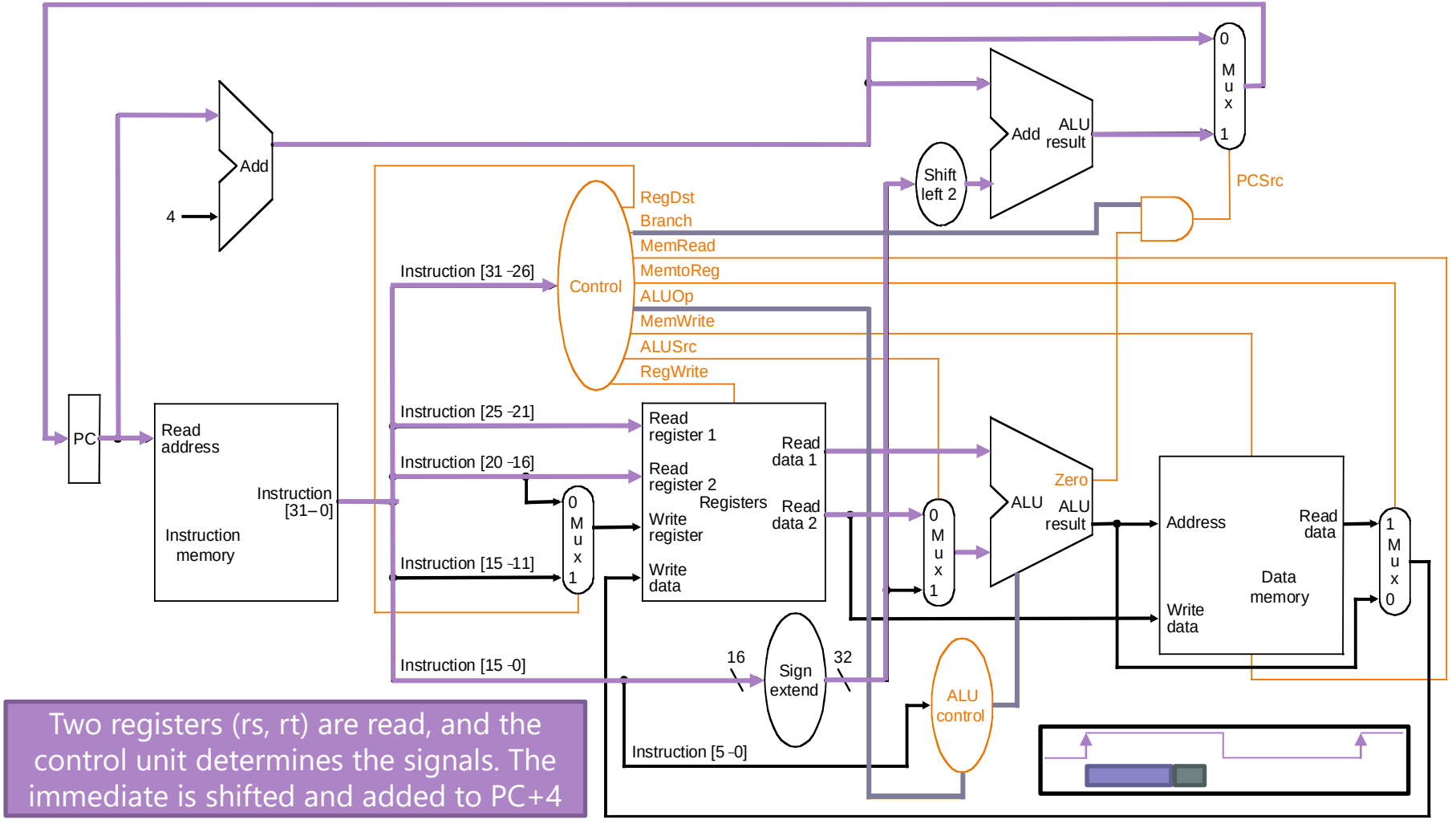
The memory data is stored in the register (rt)

Executing a beq instruction

Executing an i-type : e.g., beq

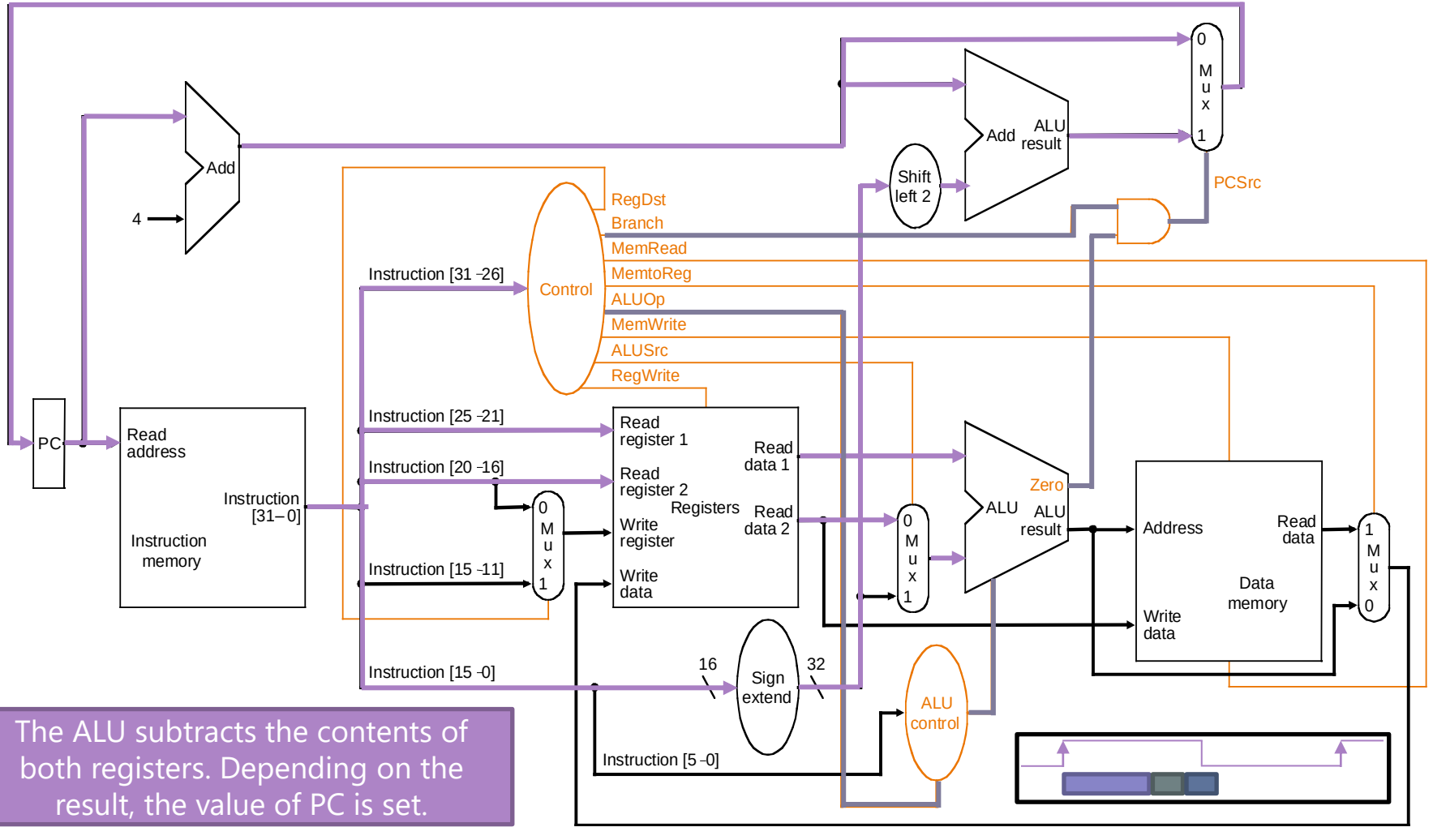


Executing an i-type : e.g., beq



Two registers (rs, rt) are read, and the control unit determines the signals. The immediate is shifted and added to PC+4

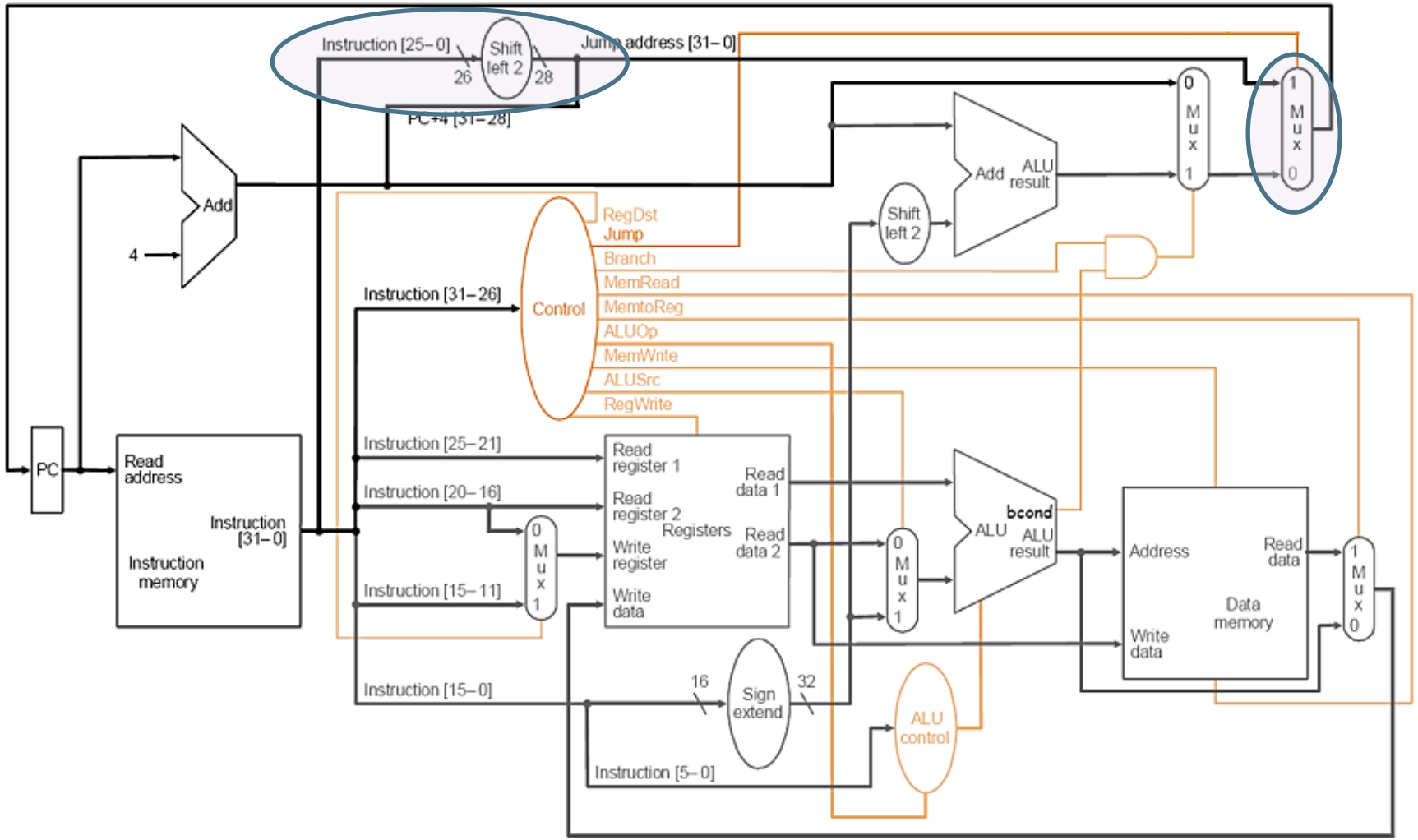
Executing an i-type : e.g., beq



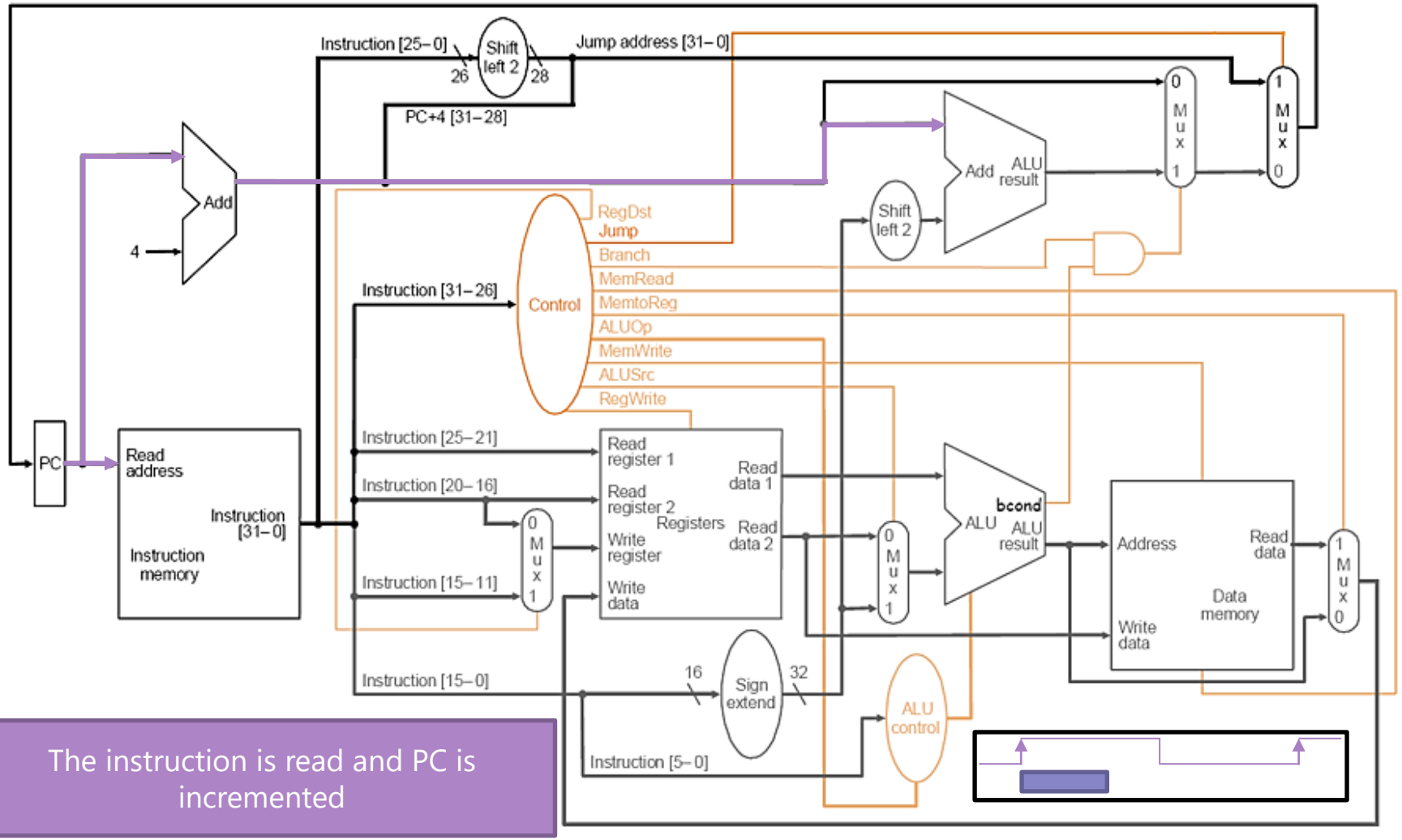
The ALU subtracts the contents of both registers. Depending on the result, the value of PC is set.

Executing a jump instruction

Extending the CPU

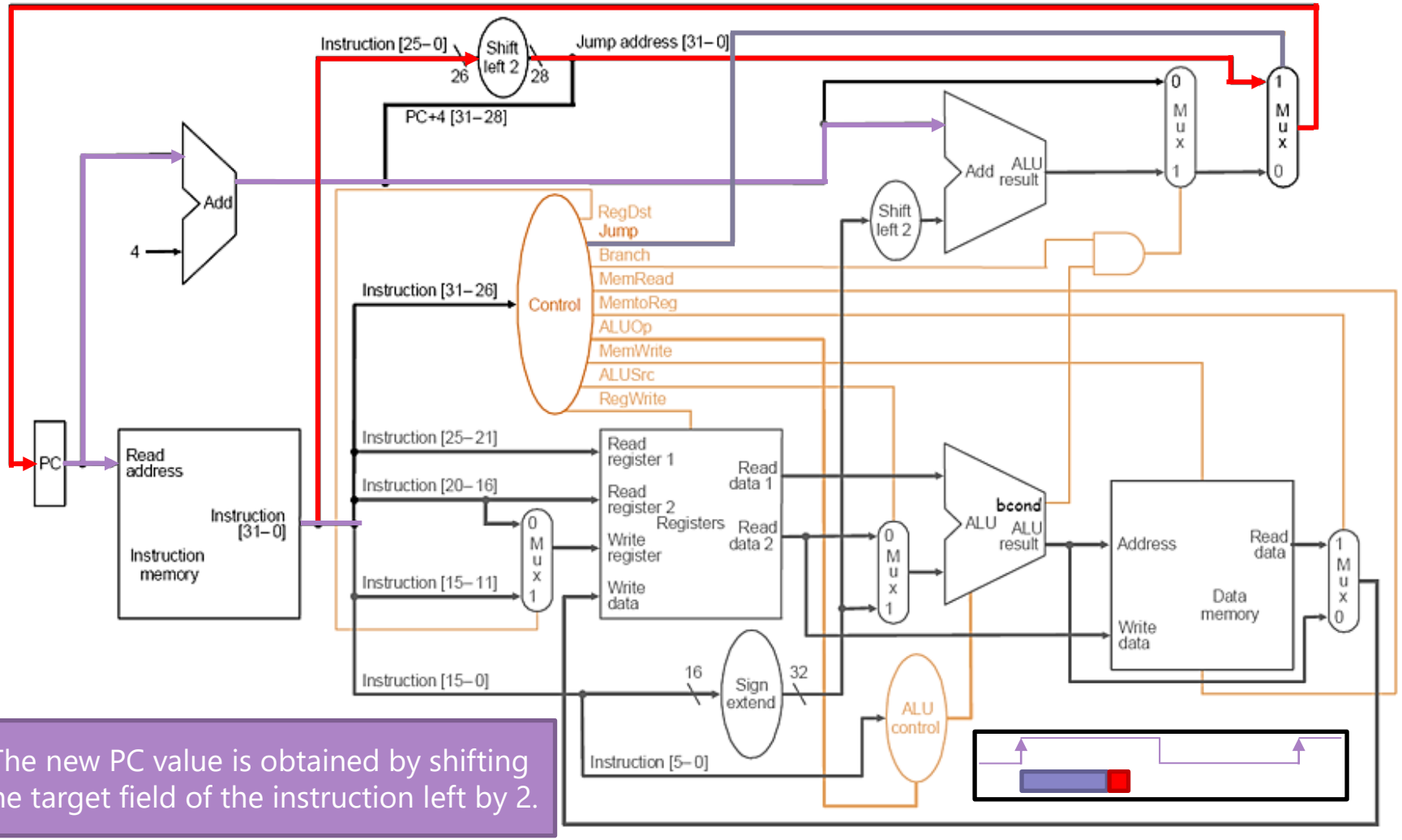


Executing an j-type



The instruction is read and PC is incremented

Executing an j-type

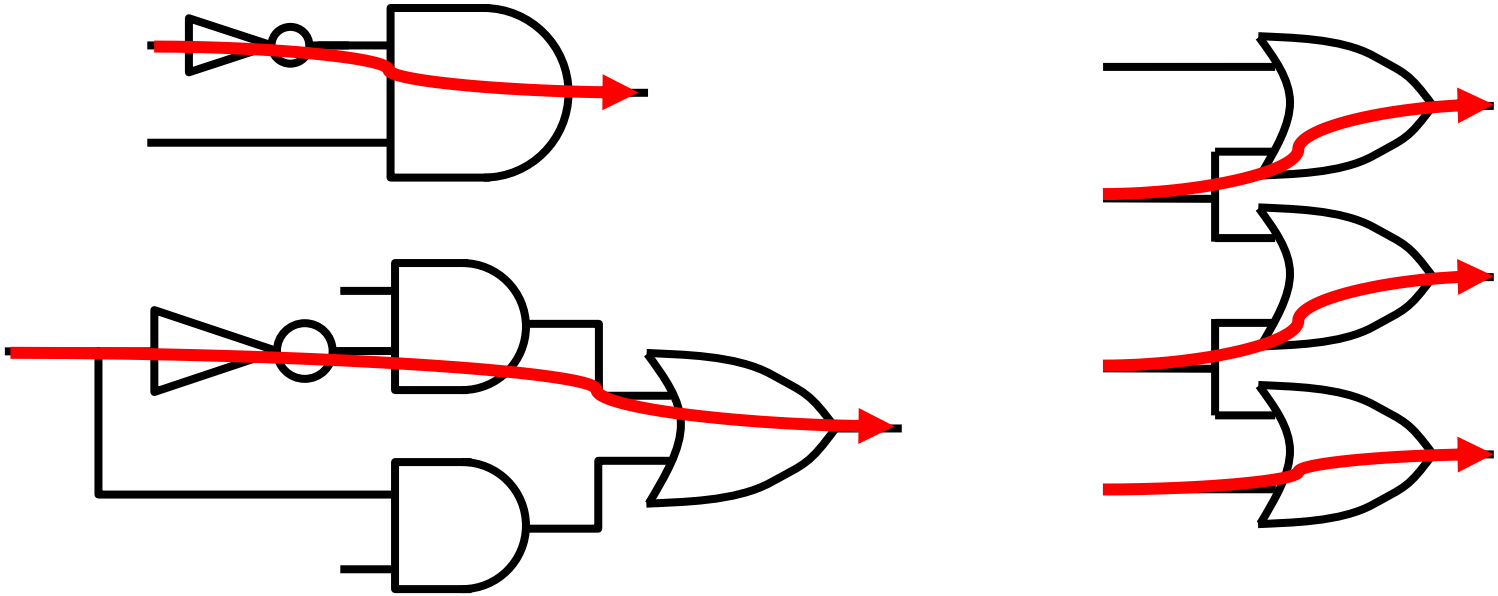


The new PC value is obtained by shifting the target field of the instruction left by 2.

Real-world clocking issues

Remember the Critical Path?

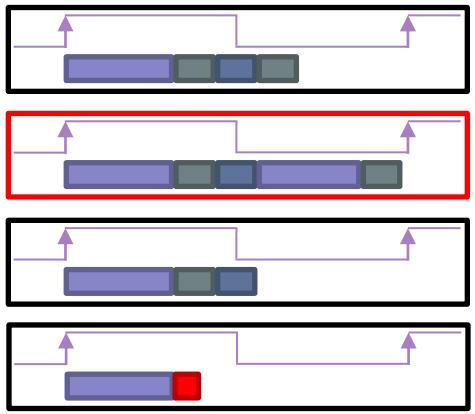
- the **critical path** is the path through a circuit that requires the longest series of **sequential** operations
  - they depend on each other and **can't be done in parallel!**

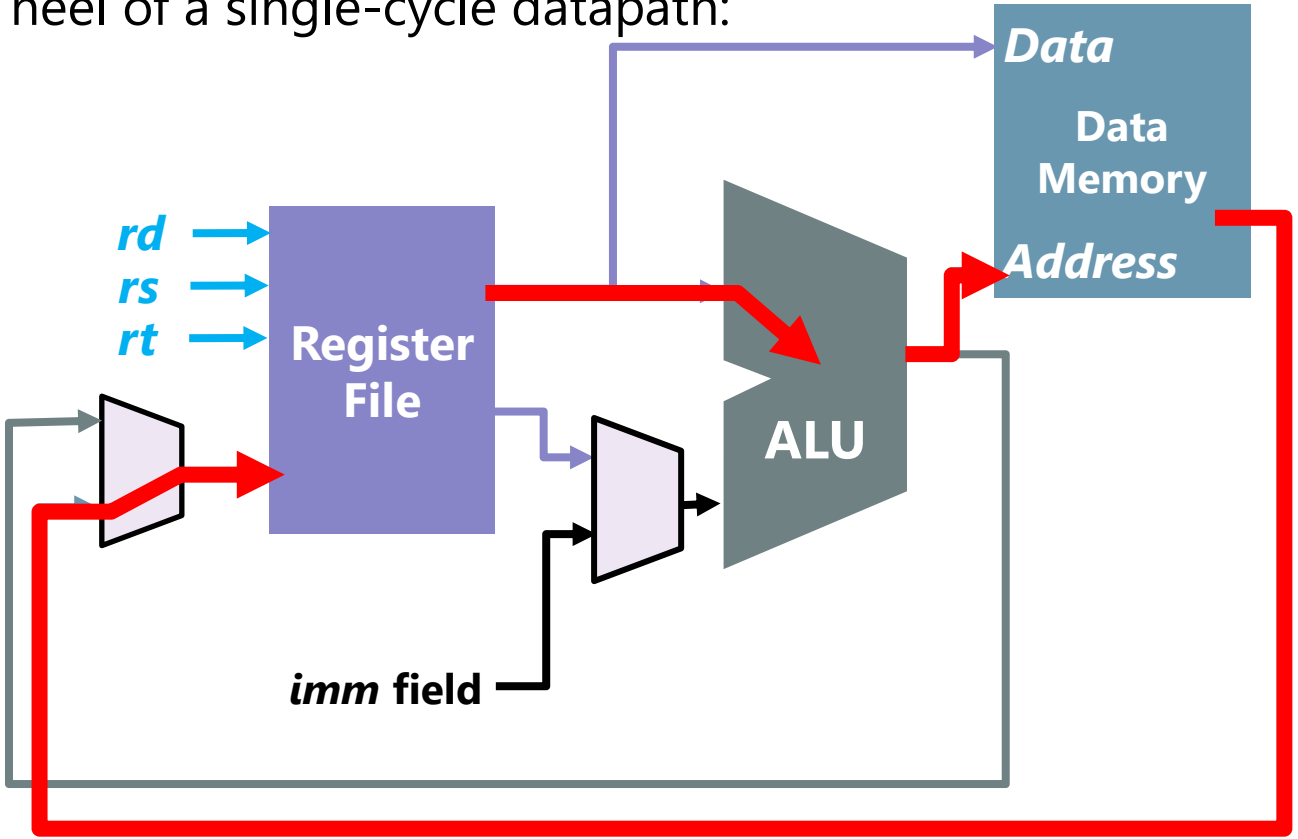


How fast can we clock it?

- what's the thing that limits the clock speed?
  - the **critical path**. in our case it happens to be...
  - and here is the Achilles' heel of a single-cycle datapath:

Memory is SLOW.



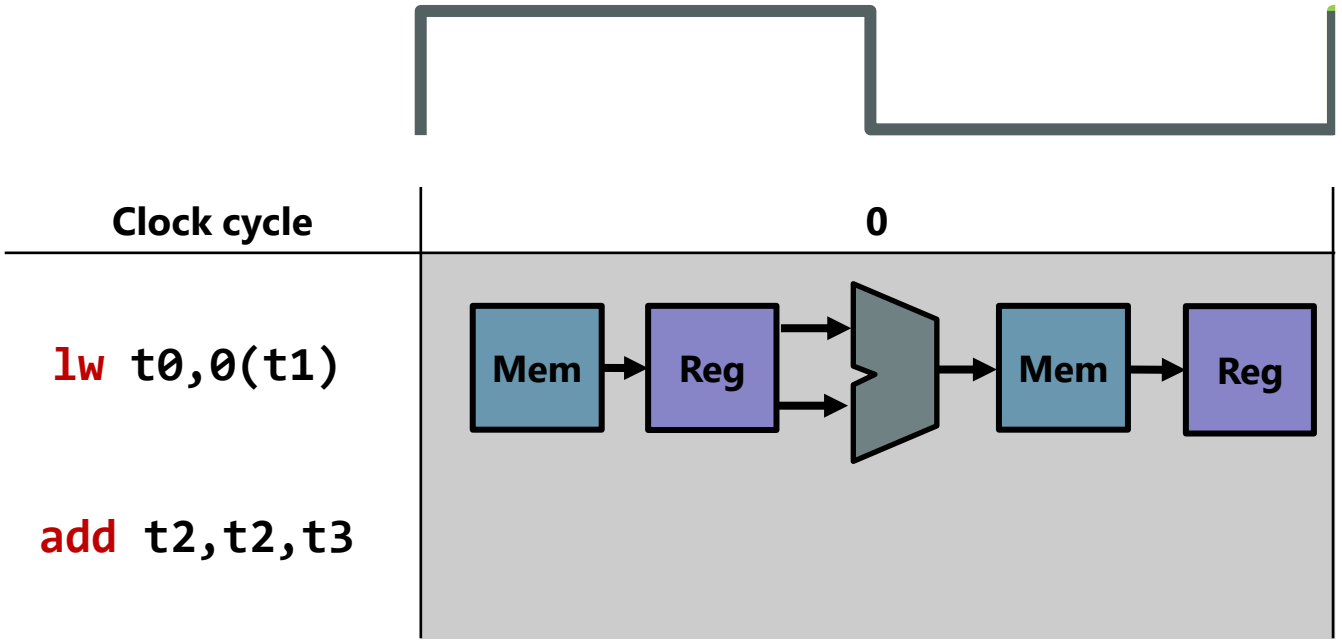


Single-cycle CPU

Any instruction executes during a single clock cycle

Length of the clock cycle must accommodate the longest instruction

- Faster instructions waste cycle time



It's bad.

- typical access times for modern DDR4 RAM is 12-15 ns
  - that's our single-cycle CPU's critical path time
- the inverse of that is... 66-83 MHz
  - **YEESH**
- the **lw** and **sw** instructions are holding us back
- all the other instructions are gonna be WAY faster

